

10 TH MATHS PUBLIC QUESTIONS 2026-2027

ONE MARK PUBLIC QUESTIONS (CHAPTER-WISE & YEAR-WISE 2019 - 2026)

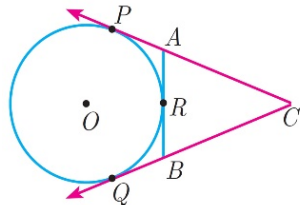
CHAPTER-1: RELATIONS AND FUNCTIONS		
S.No	Public Questions	Year
1	If $n(A \times B) = 6$ and $A = \{1,3\}$ then $n(B)$ is (A) 1 (B) 2 (C) 3 (D) 6	APR-24, JUL-21, MDL-19
2	If $A = \{1,2\}$, $B = \{1,2,3,4\}$, $C = \{5,6\}$ and $D = \{5,6,7,8\}$ then state which of the following statement is true. (A) $(A \times C) \subset (B \times D)$ (B) $(B \times D) \subset (A \times C)$ (C) $(A \times B) \subset (A \times D)$ (D) $(D \times A) \subset (B \times A)$	JUL-20
3	If the ordered pairs $(a + 2, 4)$ and $(5, 2a + b)$ are equal then (a, b) is (A) (2,-2) (B) (5,1) (C) (2,3) (D) (3,-2)	APR-22
4	If there are 1024 relations from a set $A = \{1,2,3,4,5\}$ to a set B , then the number of elements in B is (A) 3 (B) 2 (C) 4 (D) 8	JUL-22, 23, PTA-2 JUL-26
5	The range of the relation $R = \{(x, x^2) \mid x \text{ is a prime number less than } 13\}$ is (A) $\{2,3,5,7\}$ (B) $\{2,3,5,7,11\}$ (C) $\{4,9,25,49,121\}$ (D) $\{1,4,9,25,49,121\}$	JUL-22, PTA-4, APR-25
6	$A = \{a, b, p\}$, $B = \{2,3\}$, $C = \{p, q, r, s\}$ then $n[(A \cup C) \times B]$ (A) 8 (B) 20 (C) 12 (D) 16	APR-23, PTA-3, JUL-25
7	If $f: A \rightarrow B$ is a bijective function and if $n(B) = 7$, then $n(A)$ is equal to (A) 7 (B) 49 (C) 1 (D) 14	APR-24, PTA-2
8	If $\{(a, 8), (6, b)\}$ represents an identity function, then the value of a and b are respectively (A) (8,6) (B) (8,8) (C) (6,8) (D) (6,6)	JUL-24, PTA-1, MAR-26
9	If $f(x) = 2x^2$ and $g(x) = \frac{1}{3x}$, then $f \circ g$ is (A) $\frac{3}{2x^2}$ (B) $\frac{2}{3x^2}$ (C) $\frac{2}{9x^2}$ (D) $\frac{1}{6x^2}$	JUL-24 JUL-26
10	Let $n(A) = m$ and $n(B) = n$ then the total number of non-empty relations that can be defined from A to B is (A) m^n (B) n^m (C) $2^{mn} - 1$ (D) 2^{mn}	APR-25

11	Let $n(A) = p$ and $n(B) = q$ then the total number of relations that can be defined from A to B is (A) 2^p (B) 2^q (C) 2^{p+q} (D) 2^{pq}	APR-23 PTA-1
12	If $g = \{(1,1), (2,3), (3,5), (4,7)\}$ is a function given by $g(x) = \alpha x + \beta$ then the values of α and β are (A) $(-1,2)$ (B) $(2,-1)$ (C) $(-1,-2)$ (D) $(1,2)$	PTA-6, JUL-25
13	$f(x) = (x+1)^3 - (x-1)^3$ represents a function which is (A) linear (B) cubic (C) reciprocal (D) quadratic	PTA-5, MAR-26
14	Let $A = \{1,2,3,4\}$ and $B = \{4,8,9,10\}$. A function $f: A \rightarrow B$ given by $f = \{(1,4), (2,8), (3,9), (4,10)\}$ is a (A) Many-one function (B) Identity function (C) One-to-one function (D) Into function	PTA-4
15	Let f and g be two functions given by $f = \{(0,1), (2,0), (3,-4), (4,2), (5,7)\}$, $g = \{(0,2), (1,0), (2,4), (-4,2), (7,0)\}$ then the range of $f \circ g$ is (A) $\{0,2,3,4,5\}$ (B) $\{-4,1,0,2,7\}$ (C) $\{1,2,3,4,5\}$ (D) $\{0,1,2\}$	
16	Let $f(x) = 1 + x^2$ then (A) $f(xy) = f(x) \cdot f(y)$ (B) $f(xy) \geq f(x) \cdot f(y)$ (C) $f(xy) \leq f(x) \cdot f(y)$ (D) None of these	
CREATIVE PUBLIC QUESTIONS		
17	Let $f(x) = x^2 - x$, then $f(x-1) - f(x+1)$ is (A) $4x$ (B) $2 - 2x$ (C) $2 - 4x$ (D) $4x - 2$	[JUL-20]
18	Given $f(x) = (-1)^x$ is a function from $\mathbb{N}$ to $\mathbb{Z}$. Then the range of f is (A) $\{1\}$ (B) $\mathbb{N}$ (C) $\{1, -1\}$ (D) $\mathbb{Z}$	[PTA-6]
19	The given diagram represents (A) an onto function (B) a constant function (C) an one-one function (D) not a function	[PTA-3]
20	$f = \{(2,a), (3,b), (4,b), (5,c)\}$ is a (A) identity function (B) one-to-one function (C) many-one function (D) constant function	[MDL-19]
CHAPTER-2: NUMBERS AND SEQUENCES		
1	Given $F_1 = 1, F_2 = 3$ and $F_n = F_{n-1} + F_{n-2}$ then F_5 is (A) 3 (B) 5 (C) 8 (D) 11	APR-23, JUL-21,26 MDL-19
2	In an A.P., the first term is 1 and the common difference is 4. How many terms of the A.P. must be taken for their sum to be equal to 120? (A) 6 (B) 7 (C) 8 (D) 9	MDL-19

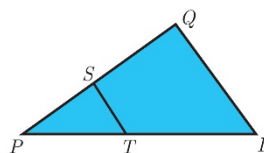
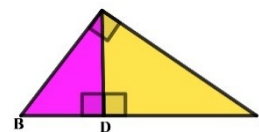
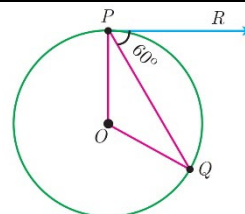
3	Using Euclid's division lemma, if the cube of any positive integer is divided by 9 then the possible remainders are (A) 0,1,8 (B) 1,4,8 (C) 0,1,3 (D) 1,3,5	PTA-5, JUL-20
4	If $A = 2^{65}$ and $B = 2^{64} + 2^{63} + 2^{62} + \dots + 2^0$ which of the following is true? (A) B is 2^{64} more than A (B) A and B are equal (C) B is larger than A by 1 (D) A is larger than B by 1	PTA-6, JUL-20
5	The sum of the exponents of the prime factors in the prime factorization of 1729 is (A) 1 (B) 2 (C) 3 (D) 4	JUL-21, 22, PTA-4
6	If the HCF of 65 and 117 is expressible in the form of $65m - 117$, then the value of m is (A) 4 (B) 2 (C) 1 (D) 3	APR-22, PTA-1, MAR-26
7	If the sequence $t_1, t_2, t_3, \dots$ are in A.P. then the sequence $t_6, t_{12}, t_{18}, \dots$ is (A) a Geometric Progression (B) an Arithmetic Progression (C) neither an Arithmetic Progression nor a Geometric Progression (D) a constant sequence	APR-23
8	$7^{4k} \equiv ___ \pmod{100}$ (A) 1 (B) 2 (C) 3 (D) 4	JUL-23, 24, PTA-1
9	The least number that is divisible by all the numbers from 1 to 10 (both inclusive) is (A) 2025 (B) 5220 (C) 5025 (D) 2520	APR-24
10	An A.P. consists of 31 terms. If its 16 th term is m , then the sum of all the terms of this A.P. is (A) $16m$ (B) $62m$ (C) $31m$ (D) $\frac{31}{2}m$	APR-24, PTA-5
11	Euclid's division lemma states that for positive integers a and b , there exist unique integers q and r such that $a = bq + r$, where r must satisfy. (A) $1 < r < b$ (B) $0 < r < b$ (C) $0 \leq r < b$ (D) $0 < r \leq b$	APR-25
12	If 6 times of 6 th term of an A.P. is equal to 7 times the 7 th term, then the 13 th term of the A.P. is (A) 0 (B) 6 (C) 7 (D) 13	APR-25, PTA-4 JUL-26
13	The value of $1^3 + 2^3 + 3^3 + \dots + 15^3 - (1 + 2 + 3 + \dots + 15)$ is (A) 14400 (B) 14200 (C) 14280 (D) 14520	JUL-25, PTA-3, MAR-26
14	The next term of the sequence $\frac{3}{16}, \frac{1}{8}, \frac{1}{12}, \frac{1}{18}, \dots$ is (A) $\frac{1}{24}$ (B) $\frac{1}{27}$ (C) $\frac{2}{3}$ (D) $\frac{1}{81}$	PTA-2
15	The first term of an arithmetic progression is unity and the common difference is 4. Which of the following will be a term of this A.P. (A) 4551 (B) 10091 (C) 7881 (D) 13531	
CREATIVE PUBLIC QUESTIONS		
16	The next term of the sequence $\frac{1}{2}, \frac{1}{6}, \frac{1}{10}, \frac{1}{14}, \dots$ is (A) $\frac{1}{15}$ (B) $\frac{1}{16}$ (C) $\frac{1}{18}$ (D) $\frac{1}{20}$	[JUL-23]

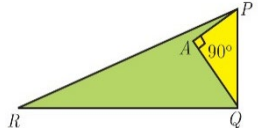
17	If t_n is the n^{th} term of an A.P., then $t_{8n} - t_n$ is (A) $(8n - 1)d$ (B) $(8n - 2)d$ (C) $(7n - 2)d$ (D) $(7nd)$	[APR-22]
18	The sequence $-3, -3, -3, \dots$ is (A) An A.P only (B) a G.P only (C) Neither A.P nor G.P (D) both A.P and G.P	[PTA-1]
19	If $2 + 4 + 6 + \dots + 2k = 90$, then the value of k is (A) 8 (B) 9 (C) 10 (D) 11	[PTA-3]
20	If a and b are two positive integers where $a > 0$ and b is a factor of a , then HCF of a and b is (A) b (B) a (C) $3ab$ (D) $\frac{a}{b}$	[PTA-4]
21	If a, b, c are in A.P then $\frac{a-b}{b-c}$ is equal to (A) $\frac{a}{b}$ (B) $\frac{b}{c}$ (C) $\frac{a}{c}$ (D) 1	[PTA-6]
CHAPTER-3: ALGEBRA		
1	The number of points of intersection of the quadratic polynomial $x^2 + 4x + 4$ with the X axis is (A) 0 (B) 1 (C) 0 or 1 (D) 2	APR-22, MDL-19
2	If A is a 2×3 matrix and B is a 3×4 matrix, how many columns does AB have (A) 3 (B) 4 (C) 2 (D) 5	MDL-19, JUL-25
3	Transpose of a column matrix is (A) unit matrix (B) diagonal matrix (C) column matrix (D) row matrix	APR-25, JUL-20, 24
4	The square root of $\frac{256x^8y^4z^{10}}{25x^6y^6z^6}$ is equal to (A) $\frac{16}{5} \left \frac{x^2z^4}{y^2} \right $ (B) $16 \left \frac{y^2}{x^2z^4} \right $ (C) $\frac{16}{5} \left \frac{y}{xz^2} \right $ (D) $\frac{16}{5} \left \frac{xz^2}{y} \right $	JUL-21
5	Graph of a linear equation is a (A) straight line (B) circle (C) parabola (D) hyperbola	APR-24, JUL-21, 23
6	If $(x - 6)$ is the HCF of $x^2 - 2x - 24$ and $x^2 - kx - 6$ then the value of k is (A) 3 (B) 5 (C) 6 (D) 8	APR-22, 25, PTA-4
7	Which of the following should be added to make $x^4 + 64$ a perfect square (A) $4x^2$ (B) $16x^2$ (C) $8x^2$ (D) $-8x^2$	APR-22, 24, PTA-2
8	A system of three linear equations in three variables is inconsistent if their planes (A) intersect only at a point (B) intersect in a line (C) coincides with each other (D) do not intersect	JUL-22, PTA-1
9	The solution of the system $x + y - 3z = -6, -7y + 7z = 7, 3z = 9$ is (A) $x = 1, y = 2, z = 3$ (B) $x = -1, y = 2, z = 3$ (C) $x = -1, y = -2, z = 3$ (D) $x = 1, y = -2, z = 3$	JUL-22, 25, MAR- 26

10	$\frac{y^2+1}{y^2}$ is not equal to (A) $\frac{y^4+1}{y^2}$ (B) $\left(y + \frac{1}{y}\right)^2$ (C) $\left(y - \frac{1}{y}\right)^2 + 2$ (D) $\left(y + \frac{1}{y}\right)^2 - 2$	JUL-22, 23, PTA-6
11	$\frac{3y-3}{y} \div \frac{7y-7}{3y^2}$ is (A) $\frac{9y}{7}$ (B) $\frac{9y^3}{21y-21}$ (C) $\frac{21y^2-42y+21}{3y^3}$ (D) $\frac{7y^2-2y+1}{y^2}$	APR-23, PTA-5
12	$\frac{x}{x^2-25} - \frac{8}{x^2+6x+5}$ gives (A) $\frac{x^2-7x+40}{(x-5)(x+5)}$ (B) $\frac{x^2+7x+40}{(x-5)(x+5)(x+1)}$ (C) $\frac{x^2-7x+40}{(x^2-25)(x+1)}$ (D) $\frac{x^2+10}{x^2-25(x+1)}$	JUL-24
13	For the given matrix $A = \begin{pmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \\ 9 & 11 & 13 & 15 \end{pmatrix}$ the order of the matrix A^T is (A) 2×3 (B) 3×2 (C) 3×4 (D) 4×3	MAR-26
14	Find the matrix X if $2X + \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix} = \begin{pmatrix} 5 & 7 \\ 9 & 5 \end{pmatrix}$ (A) $\begin{pmatrix} -2 & -2 \\ 2 & -1 \end{pmatrix}$ (B) $\begin{pmatrix} 2 & 2 \\ 2 & -1 \end{pmatrix}$ (C) $\begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}$ (D) $\begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}$	PTA-6 JUL-26
15	The solution of $(2x - 1)^2 = 9$ is equal to (A) -1 (B) 2 (C) $-1, 2$ (D) None of these	
16	The values of a and b if $4x^4 - 24x^3 + 76x^2 + ax + b$ is a perfect square are (A) 100,120 (B) 10,12 (C) $-120, 100$ (D) 12,10	
17	If the roots of the equation $q^2x^2 + p^2x + r^2 = 0$ are the squares of the roots of the equation $qx^2 + px + r = 0$, then q, p, r are in (A) A.P (B) G.P (C) Both A.P and G.P (D) none of these	
18	If number of columns and rows are not equal in a matrix then it is said to be a (A) diagonal matrix (B) rectangular matrix (C) square matrix (D) identity matrix	
19	Which of the following can be calculated from the given matrices $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, (i) A^2 (ii) B^2 (iii) AB (iv) BA (A) (i) and (ii) only (B) (ii) and (iii) only (C) (ii) and (iv) only (D) all of these	
20	If $A = \begin{pmatrix} 1 & 2 \\ 3 & 3 \\ 2 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 0 & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & 1 \\ -2 & 5 \end{pmatrix}$. Which of the following statements are correct? (A) (i) and (ii) only (B) (ii) and (iii) only (C) (iii) and (iv) only (D) all of these	

CREATIVE PUBLIC QUESTIONS		
21	The G.C.D of a^m, a^{m+1}, a^{m+2} is (A) a^m (B) a^{m+1} (C) a^{m+2} (D) 1	[JUL-21]
22	$\frac{a^2}{a^2-b^2} + \frac{b^2}{b^2-a^2} =$ _____ (A) $a - b$ (B) $a + b$ (C) $a^2 - b^2$ (D) 1	[MDL] [JUL-26]
23	The non-diagonal elements in any unit matrix are _____ (A) 0 (B) 1 (C) m (D) n	[JUL-20]
24	The LCM of $x^3 - a^3$ and $(x - a)^2$ is (A) $(x^3 - a^3)(x + a)$ (B) $(x^3 - a^3)(x - a)^2$ (C) $(x - a)^2(x^2 + ax + a^2)$ (D) $(x + a)^2(x^2 + ax + a^2)$	[PTA-1]
25	The excluded value of the rational expression $\frac{x^3+8}{x^2-2x-8}$ is (A) 8 (B) 2 (C) 4 (D) 1	[PTA-2]
26	If a polynomial is a perfect square then its factors will be repeated _____ number of times (A) odd (B) zero (C) even (D) none of the above	[PTA-4]
27	The solution of $x^2 - 25 = 0$ is (A) no real roots (B) real and equal roots (C) real and unequal roots (D) imaginary roots	[PTA-5]
28	For the given matrix $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$ the order of the matrix $(A^T)^T$ is (A) 2×3 (B) 3×2 (C) 3×4 (D) 4×3	[PTA-5]
29	On dividing $\frac{x^2-25}{x+3}$ by $\frac{x+5}{x^2-9}$ (A) $(x - 5)(x - 3)$ (B) $(x - 5)(x + 3)$ (C) $(x + 5)(x - 3)$ (D) $(x + 5)(x + 3)$	[PTA-6]
30	Graph of a quadratic equation is _____ (A) Straight line (B) Circle (C) Parabola (D) Hyperbola	[APR-23]
31	Graph of a linear polynomial is a _____ (A) Straight line (B) Circle (C) Parabola (D) Hyperbola	[PTA-2]
32	The G.C.D of $6x^2y, 9x^2yz, 12x^2y^2z$ is _____ (A) $36xy^2z^2$ (B) $36x^2y^2z$ (C) $36x^2y^2z^2$ (D) $3x^2y$	[PTA-3]
CHAPTER-4: GEOMETRY		
1	In figure CP and CQ are tangents to a circle with centre at O. ARB is another tangent touching the circle at R. If $CP = 11$ cm and $BC = 7$ cm, then the length of BR is (A) 6 cm (B) 5 cm (C) 8 cm (D) 4 cm 	MDL-19

2	In $\triangle LMN$, $\angle L = 60^\circ$, $\angle M = 50^\circ$. If $\triangle LMN \sim \triangle PQR$ then the value of $\angle R$ is (A) 40° (B) 70° (C) 30° (D) 110°	JUL-20
3	In figure if PR is tangent to the circle at P and O is the centre of the circle, then $\angle POQ$ is (A) 120° (B) 100° (C) 110° (D) 90°	JUL-20
4	If in $\triangle ABC$, $DE \parallel BC$. $AB = 3.6$ cm, $AC = 2.4$ cm and $AD = 2.1$ cm then the length of AE is (A) 1.4 cm (B) 1.8 cm (C) 1.2 cm (D) 1.05 cm	APR-24, PTA-3, JUL-21, 22, 23
5	How many tangents can be drawn to the circle from an exterior point? (A) one (B) two (C) infinite (D) zero	APR-24, JUL-21,26 22, 23
6	If $\triangle ABC$ is an isosceles triangle with $\angle C = 90^\circ$ and $AC = 5$ cm, then AB is (A) 2.5 cm (B) 5 cm (C) 10 cm (D) $5\sqrt{2}$ cm	APR-22, PTA-4
7	In a $\triangle ABC$, AD is the bisector of $\angle BAC$. If $AB = 8$ cm, $BD = 6$ cm and $DC = 3$ cm. The length of the side AC is (A) 6 cm (B) 4 cm (C) 3 cm (D) 8 cm	APR-22, PTA-6
8	If in triangles ABC and EDF , $\frac{AB}{DE} = \frac{BC}{FD}$ then they will be similar, when (A) $\angle B = \angle E$ (B) $\angle A = \angle D$ (C) $\angle B = \angle D$ (D) $\angle A = \angle F$	APR-23
9	A tangent is perpendicular to the radius at the (A) centre (B) point of contact (C) infinity (D) chord	APR-23, 25, PTA- 2, JUL-24
10	Two poles of heights 6 m and 11 m stand vertically on a plane ground. If the distance between their feet is 12 m, what is the distance between their tops? (A) 13 m (B) 14 m (C) 15 m (D) 12.8 m	JUL-24, MAR-26
11	In the adjacent figure $\angle BAC = 90^\circ$ and $AD \perp BC$ then (A) $BD \cdot CD = BC^2$ (B) $AB \cdot AC = BC^2$ (C) $BD \cdot CD = AD^2$ (D) $AB \cdot AC = AD^2$	PTA-1, JUL-25
12	In a given figure $ST \parallel QR$, $PS = 2$ cm and $SQ = 3$ cm. Then the ratio of the area of $\triangle PQR$ to the area of $\triangle PST$ is (A) 25:4 (B) 25:7 (C) 25:11 (D) 25:13	
13	The perimeters of two similar triangles $\triangle ABC$ and $\triangle PQR$ are 36 cm and 24 cm respectively. If $PQ = 10$ cm, then the length of AB is (A) $6\frac{2}{3}$ cm (B) $\frac{10\sqrt{6}}{3}$ cm (C) $66\frac{2}{3}$ cm (D) 15 cm	PTA-5



14	In the given figure, $PR = 26$ cm, $QR = 24$ cm, $\angle PAQ = 90^\circ$, $PA = 6$ cm and $QA = 8$ cm. Find $\angle PQR$ (A) 80° (B) 85° (C) 75° (D) 90°		
15	The two tangents from an external points P to a circle with centre at O are PA and PB . If $\angle APB = 70^\circ$ then the value of $\angle AOB$ is (A) 100° (B) 110° (C) 120° (D) 130°		
CHAPTER-5: CO-ORDINATE GEOMETRY			
1	The slope of the line joining $(12,3)$, $(4,a)$ is $\frac{1}{8}$. The value of ' a ' is (A) 1 (B) 4 (C) -5 (D) 2		MDL-19, PTA-3 JUL-26
2	The straight line given by the equation $x = 11$ is (A) parallel to X axis (B) parallel to Y axis (C) passing through the origin (D) passing through the point $(0,11)$		JUL-20, 23, 24, PTA-1, MAR-26
3	The area of triangle formed by the points $(-5,0)$, $(0,-5)$ and $(5,0)$ is (A) 0 sq.units (B) 25 sq.units (C) 5 sq.units (D) none of these		JUL-21, 25, PTA- 2, APR-24
4	If $(5,7)$, $(3,p)$ and $(6,6)$ are collinear, then the value of p is (A) 3 (B) 6 (C) 9 (D) 12		PTA-5, APR-22
5	The point of intersection of the straight line $3x - y = 4$ and $x + y = 8$ is (A) $(5,3)$ (B) $(2,4)$ (C) $(3,5)$ (D) $(4,4)$		JUL-22, PTA-2, MAR-26
6	The slope of the line which is perpendicular to a line joining the points $(0,0)$ and $(-8,8)$ is (A) -1 (B) 1 (C) $\frac{1}{3}$ (D) -8		APR-22, 25
7	If slope of the line PQ is $\frac{1}{\sqrt{3}}$ then slope of the perpendicular bisector of PQ is (A) $\sqrt{3}$ (B) $-\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) 0		JUL-22, 23, PTA-6
8	The equation of a line passing through the origin and perpendicular to the line $7x - 3y + 4 = 0$ is (A) $7x - 3y + 4 = 0$ (B) $3x - 7y + 4 = 0$ (C) $3x + 7y = 0$ (D) $7x - 3y = 0$		JUL-24, PTA-4
9	When proving that a quadrilateral is a trapezium, it is necessary to show (A) Two sides are parallel. (B) Two parallel and two non-parallel sides. (C) Opposite sides are parallel. (D) All sides are of equal length.		PTA-4, JUL-25
10	A man walks near a wall, such that the distance between him and the wall is 10 units. Consider the wall to be the Y axis. The path travelled by the man is (A) $x = 10$ (B) $y = 10$ (C) $x = 0$ (D) $y = 0$		JUL-26

11	If A is a point on the Y axis whose ordinate is 8 and B is a point on the X axis whose abscissae is 5 then the equation of the line AB is (A) $8x + 5y = 40$ (B) $8x - 5y = 40$ (C) $x = 8$ (D) $y = 5$	
12	Consider four straight lines (i) $l_1; 3y = 4x + 5$ (ii) $l_2; 4y = 3x - 1$ (iii) $l_3; 4y + 3x = 7$ (iv) $l_4; 4x + 3y = 2$ Which of the following statement is true? (A) l_1 and l_2 are perpendicular (B) l_1 and l_4 are parallel (C) l_2 and l_4 are perpendicular (D) l_2 and l_3 are parallel	
13	A straight line has equation $8y = 4x + 21$. Which of the following is true (A) The slope is 0.5 and the y intercept is 2.6 (B) The slope is 5 and the y intercept is 1.6 (C) The slope is 0.5 and the y intercept is 1.6 (D) The slope is 5 and the y intercept is 2.6	PTA-3
14	When proving that a quadrilateral is a parallelogram by using slopes you must find (A) The slopes of two sides (B) The slopes of two pair of opposite sides (C) The lengths of all sides (D) Both the lengths and slopes of two sides	
15	$(2,1)$ is the point of intersection of two lines. (A) $x - y - 3 = 0; 3x - y - 7 = 0$ (B) $x + y = 3; 3x + y = 7$ (C) $3x + y = 3; x + y = 7$ (D) $x + 3y - 3 = 0; x - y - 7 = 0$	PTA-3
CREATIVE PUBLIC QUESTIONS		
16	The equation of a line passing through the origin and parallel to the line $7x - 3y + 4 = 0$ is (A) $7x - 3y + 4 = 0$ (B) $3x - 7y + 4 = 0$ (C) $3x + 7y = 0$ (D) $7x - 3y = 0$	[APR-25, APR-23]
17	The slope of the straight line perpendicular to x -axis is (A) 1 (B) 0 (C) ∞ (D) -1	[JUL-21]
18	The perimeter of a triangle formed by the points $(0,0)$, $(1,0)$ and $(0,1)$ is (A) $\sqrt{2}$ (B) 2 (C) $2 + \sqrt{2}$ (D) $2 - \sqrt{2}$	[PTA-5]
19	If the points $A(6,1)$, $B(8,2)$, $C(9,4)$ and $D(p, 3)$ are the vertices of a parallelogram, taken in order then the value of p is (A) -7 (B) 7 (C) 6 (D) -6	[PTA-5]
CHAPTER-6: TRIGONOMETRY		
1	If $x = a \tan \theta$ and $y = b \sec \theta$ then (A) $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (C) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (D) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$	APR-24, MDL-19 JUL-26
2	If the ratio of the height of a tower and the length of its shadow is $\sqrt{3}:1$, then the angle of elevation of the sun has measure (A) 45° (B) 30° (C) 90° (D) 60°	APR-25, JUL-21, 24, PTA-6

3	A tower is 60 m high. Its shadow reduces by x metres when the angle of elevation of the sun increases from 30° to 45° then x is equal to (A) 41.92 m (B) 43.92 m (C) 43 m (D) 45.6 m	APR-22
4	The angle of elevation of a cloud from a point h metres above a lake is β . The angle of depression of its reflection in the lake is 45° . The height of location of the cloud from the lake is (A) $\frac{h(1+\tan\beta)}{1-\tan\beta}$ (B) $\frac{h(1-\tan\beta)}{1+\tan\beta}$ (C) $h\tan(45^\circ - \beta)$ (D) none of these	JUL-22
5	If $\sin\theta = \cos\theta$, then $2\tan^2\theta + \sin^2\theta - 1$ is equal to (A) $-\frac{3}{2}$ (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $-\frac{2}{3}$	APR-23, PTA-4
6	$\frac{\tan\theta}{\csc^2\theta - \tan\theta}$ is equal to (A) $\sec\theta$ (B) $\cot^2\theta$ (C) $\sin\theta$ (D) $\cot\theta$	JUL-23, 24, PTA-3
7	The value of $\frac{\sin^2\theta+1}{1+\tan^2\theta}$ is equal to (A) $\tan^2\theta$ (B) 1 (C) $\cot^2\theta$ (D) 0	JUL-25
8	If $(\sin\alpha + \csc\alpha)^2 + (\cos\alpha + \sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, then the value of k is equal to (A) 9 (B) 7 (C) 5 (D) 3	PTA-1
9	If $\sin\theta + \cos\theta = a$ and $\sec\theta + \csc\theta = b$, then the value of $b(a^2 - 1)$ is equal to (A) $2a$ (B) $3a$ (C) 0 (D) $2ab$	
10	If $5x = \sec\theta$ and $\frac{5}{y} = \tan\theta$, then $x^2 - \frac{1}{y^2}$ is equal to (A) 25 (B) $\frac{1}{25}$ (C) 5 (D) 1	PTA-2
11	$(1 + \tan\theta + \sec\theta)(1 + \cot\theta - \csc\theta)$ is equal to (A) 0 (B) 1 (C) 2 (D) -1	
12	$a\cot\theta + b\csc\theta = p$ and $b\cot\theta + a\csc\theta = q$ then $p^2 - q^2$ is equal to (A) $a^2 - b^2$ (B) $b^2 - a^2$ (C) $a^2 + b^2$ (D) $b - a$	PTA-5
13	The electric pole subtends an angle of 30° at a point on the same level as its foot. At a second point ' b ' metres above the first, the depression of the foot of the pole is 60° . The height of the pole (in metres) is equal to (A) $\sqrt{3}b$ (B) $\frac{b}{3}$ (C) $\frac{b}{2}$ (D) $\frac{b}{\sqrt{3}}$	
14	The angle of depression of the top and bottom of 20 m tall building from the top of a multistoried building are 30° and 60° respectively. The height of the multistoried building and the distance between two buildings (in metres) is (A) $20, 10\sqrt{3}$ (B) $30, 5\sqrt{3}$ (C) 20, 10 (D) $30, 10\sqrt{3}$	
15	Two persons are standing ' x ' metres apart from each other and the height of the first person is double that of the other. If from the middle point of the line joining their feet an observer finds the angular elevations of their tops to be complementary, then the height of the shorter person (in metres) is (A) $\sqrt{2}x$ (B) $\frac{x}{2\sqrt{2}}$ (C) $\frac{x}{\sqrt{2}}$ (D) $2x$	

CREATIVE PUBLIC QUESTIONS		
16	If $\tan\theta + \cot\theta = 2$ then $\tan^2\theta + \cot^2\theta$ is equal to (A) 0 (B) 1 (C) 2 (D) 4	[JUL-20, MAR-26]
17	The angle of elevation and depression are usually measured by a device called (A) Clinometer (B) kaleidoscope (C) Periscope (D) Telescope	[PTA-1]
18	$\frac{\sin(90^\circ - \theta)\sin\theta}{\tan\theta} + \frac{\cos(90^\circ - \theta)\cos\theta}{\cot\theta} =$ (A) $\tan\theta$ (B) 1 (C) -1 (D) $\sin\theta$	[PTA-2]
CHAPTER-7: MENSURATION		
1	The height of a right circular cone whose radius is 5 cm and slant height is 13 cm will be (A) 12 cm (B) 10 cm (C) 13 cm (D) 5 cm	APR-23, JUL-21, 24, MDL-19
2	A spherical ball of radius r_1 units is melted to make 8 new identical balls each of radius r_2 units. Then $r_1:r_2$ is (A) 2:1 (B) 1:2 (C) 4:1 (D) 1:4	JUL-20, PTA-6
3	The total surface area of a hemi-sphere is how much times the square of its radius. (A) π (B) 4π (C) 3π (D) 2π	JUL-21, 22, 23, PTA-3
4	If two solid hemispheres of same base radius r units are joined together along their bases, then curved surface area of this new solid is (A) $4\pi r^2$ sq. units (B) $6\pi r^2$ sq. units (C) $3\pi r^2$ sq. units (D) $8\pi r^2$ sq. units	APR-22
5	If the radius of the base of a right circular cylinder is halved keeping the same height, then the ratio of the volume of the cylinder thus obtained to the volume of original cylinder is (A) 1:2 (B) 1:4 (C) 1:6 (D) 1:8	JUL-22
6	The ratio of the volumes of a cylinder, a cone and a sphere, if each has the same diameter and same height is (A) 1:2:3 (B) 2:1:3 (C) 1:3:2 (D) 3:1:2	APR-23, 24, PTA-5, JUL-25
7	The curved surface area of a right circular cone of height 15 cm and base diameter 16 cm is (A) $60\pi \text{ cm}^2$ (B) $68\pi \text{ cm}^2$ (C) $120\pi \text{ cm}^2$ (D) $136\pi \text{ cm}^2$	APR-25, JUL-23, 25
8	The volume (in cm^3) of the greatest sphere that can be cut off from a cylindrical log of wood of base radius 1 cm and height 5 cm is (A) $\frac{4}{3}\pi$ (B) $\frac{10}{3}\pi$ (C) 5π (D) $\frac{20}{3}\pi$	APR-25
9	The total surface area of a cylinder whose radius is $\frac{1}{3}$ of its height is (A) $\frac{9\pi h^2}{8}$ sq.units (B) $24\pi h^2$ sq.units (C) $\frac{8\pi h^2}{9}$ sq.units (D) $\frac{56\pi h^2}{9}$ sq.units	PTA-1 JUL-26

10	The height and radius of the cone of which the frustum is a part are h_1 units and r_1 units respectively. Height of the frustum is h_2 units and radius of the smaller base is r_2 units. If $h_2:h_1 = 1:2$ then $r_2:r_1$ is (A) 1:3 (B) 1:2 (C) 2:1 (D) 3:1	PTA-2
11	In a hollow cylinder, the sum of the external and internal radii is 14 cm and the width is 4 cm. If its height is 20 cm, the volume of the material in it is (A) $5600\pi \text{ cm}^3$ (B) $1120\pi \text{ cm}^3$ (C) $56\pi \text{ cm}^3$ (D) $3600\pi \text{ cm}^3$	PTA-4
12	If the radius of the base of a cone is tripled and the height is doubled then the volume is (A) made 6 times (B) made 18 times (C) made 12 times (D) unchanged	
13	A solid sphere of radius x cm is melted and cast into a shape of a solid cone of same radius. The height of the cone is (A) $3x$ cm (B) x cm (C) $4x$ cm (D) $2x$ cm	
14	A frustum of a right circular cone is of height 16 cm with radii of its ends as 8 cm and 20 cm. Then, the volume of the frustum is (A) $3328\pi \text{ cm}^3$ (B) $3228\pi \text{ cm}^3$ (C) $3240\pi \text{ cm}^3$ (D) $3340\pi \text{ cm}^3$	
15	A shuttle cock used for playing badminton has the shape of the combination of (A) a cylinder and a sphere (B) a hemisphere and a cone (C) a sphere and a cone (D) frustum of a cone and a hemisphere	
CREATIVE PUBLIC QUESTIONS		
16	The curved surface area of a right circular Cylinder of height 4 cm and base diameter 10 cm is (A) 40π sq.units (B) 20π sq.units (C) 14π sq.units (D) 80π sq.units	[APR-24]
17	If the radius of the cylinder is doubled, the new volume of the cylinder will be ____ times the original volume. (A) Same (B) 3 (C) 4 (D) 2	[JUL-22]
18	C.S.A of solid sphere is equal to (A) T.S.A of solid sphere (B) T.S.A of hemisphere (C) C.S.A of hemisphere (D) none of these	[PTA-5]
19	A child reshapes a cone made up of clay of height 24 cm and radius 6 cm into a sphere, then the radius of sphere is (A) 24 cm (B) 12 cm (C) 6 cm (D) 48 cm	[JUL-20]
20	If the volume of sphere is $36\pi \text{ cm}^3$, then its radius is equal to (A) 3 cm (B) 2 cm (C) 5 cm (D) 10 cm	[PTA-3]
21	If the volume of solid hemisphere is numerically equal to its total surface area then the diameter is: (A) $\frac{9}{2}$ units (B) 9 units (C) 3π units (D) $\frac{3}{2}$ units	JUL-26

CHAPTER-8: STATISTICS AND PROBABILITY		
1	Variance of first 20 natural numbers is (A) 32.25 (B) 44.25 (C) 33.25 (D) 30	PTA-5, MDL-19
2	If the mean and coefficient of variation of a data are 4 and 87.5% then the standard deviation is (A) 3.5 (B) 3 (C) 4.5 (D) 2.5	MDL-19
3	The mean of 100 observations is 40 and their standard deviation is 3. The sum of squares of all observations is (A) 40000 (B) 160900 (C) 160000 (D) 30000	JUL-20
4	If a letter is chosen at random from the English alphabets $\{a, b, \dots, z\}$, then the probability that the letter chosen precedes x (A) $\frac{12}{13}$ (B) $\frac{1}{13}$ (C) $\frac{23}{26}$ (D) $\frac{3}{26}$	APR-23, JUL-20, PTA-3
5	A page is selected at random from a book. The probability that the digit at units place of the page number chosen is less than 7 is (A) $\frac{3}{10}$ (B) $\frac{7}{10}$ (C) $\frac{3}{9}$ (D) $\frac{7}{9}$	JUL-21, 22, 25 JUL-26
6	The probability of getting a job for a person is $\frac{x}{3}$. If the probability of not getting the job is $\frac{2}{3}$ then the value of x is (A) 2 (B) 1 (C) 3 (D) 1.5	APR-22, 24
7	The range of the data 8,8,8,8...8 is (A) 0 (B) 1 (C) 8 (D) 3	JUL-23, 24, PTA-3
8	The probability a red marble selected at random from a jar containing p red, q blue and r green marbles is (A) $\frac{q}{p+q+r}$ (B) $\frac{p}{p+q+r}$ (C) $\frac{p+q}{p+q+r}$ (D) $\frac{p+r}{p+q+r}$	JUL-23, MAR-26
9	Which of the following is incorrect? (A) $P(A) > 1$ (B) $0 \leq P(A) \leq 1$ (C) $P(\phi) = 0$ (D) $P(A) + P(\bar{A}) = 1$	APR-25, JUL-24, PTA-1, 4, 5
10	A purse contains 10 notes of , 15 notes of , and 25 notes of . One note is drawn at random. What is the probability that the note is either a note or note? (A) $\frac{1}{5}$ (B) $\frac{3}{10}$ (C) $\frac{2}{3}$ (D) $\frac{4}{5}$	APR-25
11	Which of the following is not a measure of dispersion? (A) Range (B) Standard deviation (C) Arithmetic mean (D) Variance	PTA-6
12	The sum of all deviations of the data from its mean is (A) Always positive (B) always negative (C) zero (D) non-zero integer	PTA-1 JUL-26
13	The standard deviation of a data is 3. If each value is multiplied by 5 then the new variance is (A) 3 (B) 15 (C) 5 (D) 225	

14	If the standard deviation of x, y, z is p then the standard deviation of $3x + 5, 3y + 5, 3z + 5$ is (A) $3p + 5$ (B) $3p$ (C) $p + 5$ (D) $9p + 15$	
15	Kamalam went to play a lucky draw contest. 135 tickets of the lucky draw were sold. If the probability of Kamalam winning is $\frac{1}{9}$, then the number of tickets bought by Kamalam is (A) 5 (B) 10 (C) 15 (D) 20	
CREATIVE PUBLIC QUESTIONS		
16	Which of the following values cannot be a probability of an event? (A) 0 (B) 0.5 (C) 1.05 (D) 1	[APR-24]
17	If the sum and mean of a data are 407 and 11 respectively, then the number of observations in the data are: (A) 37 (B) 4477 (C) 396 (D) 418	[APR-23]
18	A fair die is thrown once. The probability of getting a prime (or) composite number is (A) 1 (B) 0 (C) $\frac{5}{6}$ (D) $\frac{1}{6}$	[PTA-6]
19	For a collection of data, $\Sigma x^2 = 140$, $\Sigma x = 28$ and $n = 7$, the standard deviation is (A) 4 (B) 2 (C) 16 (D) 8	[MAR-26]